add all p's neighbors within radius ε to the same cluster as p, and hence w will end up in the same cluster as p.

What if there are two core points p and q, that are density reachable from each other, and both reachable from w? Well, by definition, there will be a chain of core points w1, …, wn between q and p, and so in turn each core point in the path will be added to the same cluster as q, and finally so will p as well, when it's wn's turn.

What if p and q are core points but not reachable from each other, but both reachable from a point w? Can w be a core point?

Let's reason *ad absurdum*[14]: suppose that w is reachable from p, a core point, that is processed before w. Hence there is a chain of core points p1, …, pn, each reachable from the previous one, that connects p to w.

Suppose also that w, by the time pn is processed, has already been added to another cluster, different from p's: this means that there is a core point q, for which w is reachable from q (and hence there is a chain of core points q1, …, qk etc…), but p is not reachable from q, reason for which they are in different clusters.

Now, w can be a core point, or not a core point.

If w was a core point, then by definition there would be a chain made of the core points q, q1, …, qk, w, p1, …, pn, p, where all points are reachable from each other; therefore, p